
A Theory of Simulation for Language Models: Persona Conditioning Broadens Behavior but Does Not Recover Human Outcome Tails

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Abstract

Large language models are increasingly used as stand-ins for people in product evaluation, social simulation, and agent design. The central question is not whether these models can produce human-like text, but whether they can simulate the distribution of human outcomes once we condition them on latent factors such as demographics or persona. We propose a simple theory of simulation with three requirements: local fidelity to known human-style judgments, responsiveness to latent factors, and coverage of plausible behavioral variation. We test this framework with real API calls to GPT-4.1 on May 5, 2026 using two tasks: a 90-item SOCIAL IQA social-judgment benchmark and a Turing-Experiment-style ultimatum game with 15 synthetic participants per condition. We compare unconditioned, demographic-conditioned, and persona-conditioned prompting while holding the base model fixed.

Persona conditioning improves SOCIAL IQA accuracy from 0.744 to 0.789 and raises the number of distinct ultimatum policies from 2 to 5. However, all conditions still accept unfair offers at unrealistically high rates, with mean acceptance thresholds between 2.40 and 2.67 on a 1–9 offer scale. The added diversity therefore broadens behavior only within a narrow cooperative family. Our main conclusion is that current frontier LLMs are controllable but compressed behavioral simulators: they can support local fidelity and some latent-factor responsiveness, but they do not yet simulate a true superset of human outcomes. This distinction matters for any application that relies on subgroup structure or tail behavior.

1 Introduction

Large language models are already being used as synthetic users, survey respondents, and behavioral stand-ins. That trend raises a narrow but important question: when does a model simulate people, rather than merely produce plausible text about people? If an LLM collapses toward an articulate, cooperative, average-seeming participant, then evaluations built on top of it will miss the disagreement, irrationality, and subgroup structure that often matter most.

Prior work shows that LLMs can replicate some human-subject findings, express persona-like differences, and support believable agent interactions Park et al. [2023], Aher et al. [2023], Xie et al. [2024], Sorokovikova et al. [2024]. At the same time, several studies argue that these successes can be misleading. Population alignment remains uneven across demographic groups Santurkar et al. [2023], models often look more rational than real people Liu et al. [2025], and long-horizon simulation benchmarks expose strong homogenization and positivity biases Chen et al. [2026]. The missing piece is a theory of simulation that distinguishes stylistic plausibility from distributional validity.

What is a useful theory of simulation? We define simulation with three requirements. First, a simulator should show *local fidelity*: it should match known human-style judgments on benchmarked

tasks. Second, it should show *latent-factor responsiveness*: changing relevant conditioning variables such as persona or demographics should change behavior in coherent ways. Third, it should show *coverage*: the simulator should preserve a plausible range of outcomes rather than collapsing onto a narrow average. This last requirement is the hardest one, and it is the one most directly tied to the claim that LLMs can simulate a superset of human outcomes.

We study this question with one model family and two complementary tasks. We query GPT-4.1 through the chat completions API under three prompt conditions: unconditioned, demographic-conditioned, and persona-conditioned. We then evaluate a labeled social reasoning benchmark, SOCIAL IQA, and a Turing-Experiment-style ultimatum game adapted from local replication assets Aher et al. [2023]. The first task measures local judgment fidelity. The second probes distributional behavior by asking whether the model produces distinct acceptance policies once latent factors are varied.

Our experiments point to a middle position. Persona conditioning improves SOCIAL IQA accuracy from 0.744 to 0.789, a 4.4-point gain over the unconditioned baseline and an 8.9-point gain over demographic conditioning. It also expands observed ultimatum policies from 2 to 5 distinct profiles. However, all three conditions accept unfair offers much too early: mean acceptance thresholds remain between 2.40 and 2.67 on a 1–9 offer scale, and every condition accepts offers of 3 or higher almost universally. These results suggest that conditioning broadens the behavioral manifold without recovering human-like tails.

We make four contributions:

- We propose a simple theory of LLM simulation based on local fidelity, latent-factor responsiveness, and coverage.
- We conduct two real-API experiments on GPT-4.1 that separate pointwise social-judgment accuracy from distributional game behavior.
- We show that persona conditioning can improve fidelity and widen support at the same time, but the added variation remains compressed within a cooperative policy family.
- We argue that LLM simulators should be validated separately for fidelity, subgroup structure, and tail coverage rather than treated as drop-in substitutes for human populations.

The rest of the paper is organized as follows. Section 2 situates our study within prior work on human simulation and behavioral alignment. Section 3 describes the tasks, prompting conditions, and metrics. Section 4 reports the main empirical findings, and Section 5 discusses their theoretical implications and limitations.

2 Related Work

Believable simulation is not the same as distributional validity. Generative agent work shows that LLM-based systems can produce coherent, believable behavior in interactive environments Park et al. [2023]. That line of work is important because it demonstrates process-level plausibility and coordination. However, believable behavior in a sandbox does not establish that the model captures the empirical distribution of human outcomes. Our focus is narrower: we ask whether conditioning can widen behavior while preserving fidelity to known regularities.

Human-study replication motivates our task design. Aher et al. introduce Turing Experiments as a way to test whether language models can reproduce established findings from behavioral studies Aher et al. [2023]. Xie et al. show that GPT-4-class agents can align with some human trust-game regularities when equipped with careful prompting and agent structure Xie et al. [2024]. We build on this evaluation tradition, but we focus on two distinct levels of evidence: benchmarked social judgment and population-style policy variation. Unlike prior work that treats successful replication on one task as broad evidence of simulation, we explicitly separate local fidelity from coverage.

Population alignment remains a central challenge. Santurkar et al. demonstrate that demographic steering does not reliably recover the opinion distributions of real demographic groups Santurkar et al. [2023]. Lin argues that LLMs should be treated as pragmatic simulation tools rather than replacements for human participants, and warns against reading alignment as explanation Lin [2025]. Our work operationalizes this conceptual caution. We ask not only whether the model looks plausi-

ble, but whether latent conditioning changes the response distribution in a way that supports stronger simulation claims.

Recent evidence points to collapse toward an average, rationalized person. Several studies now report systematic distortions in LLM-based human simulation. Sorokovikova et al. find that models can express personality-like variation, but the realism and stability of these traits remain open Sorokovikova et al. [2024]. Liu et al. show that strong LLMs assume people are more rational than they really are Liu et al. [2025]. Chen et al. report that long-horizon simulation benchmarks reveal hyper-activity, persona homogenization, and utopian bias Chen et al. [2026]. Our results fit this picture: conditioning increases variety, but the variation remains constrained inside a narrow and overly cooperative family of responses.

3 Methodology

Simulation criteria. We evaluate LLM simulation through three requirements. *Local fidelity* measures whether the model matches known human-style answers on benchmarked tasks. *Latent-factor responsiveness* measures whether meaningful conditioning variables, such as persona or demographics, produce coherent behavioral differences. *Coverage* measures whether the resulting behaviors span a plausible range of outcomes rather than collapsing toward one default policy. We use one task for local fidelity and one for distributional coverage because a single benchmark cannot establish all three requirements.

3.1 Model and Prompt Conditions

We use GPT-4.1 as the sole simulator, queried through the ‘v1/chat/completions’ endpoint on May 5, 2026 OpenAI [2026a]. This design keeps model capacity fixed so that differences can be attributed to conditioning rather than architecture. For SOCIAL IQA, we use temperature 0.8 and a 120-token completion cap. For the ultimatum game, we use temperature 0.9 and the same token cap. The evaluation harness uses deterministic integer seeds per condition and item or participant, as implemented in the local runner.

We compare three prompting conditions:

- UNCONDITIONED: a generic plausible U.S. adult participant.
- DEMOGRAPHIC: one of four demographic role prompts, cycled across participants.
- PERSONA: one of 15 concise persona profiles with four stable traits each.

3.2 Tasks and Data

SOCIAL IQA fidelity benchmark. We sample 90 roughly label-balanced validation examples from the local SOCIAL IQA mirror. For each item, the model sees a short scenario, a question, three answer choices, and a role prompt describing the simulated participant. The target metric is exact-choice accuracy against the dataset label, along with macro-F1 and answer entropy.

Ultimatum-game distribution benchmark. We adapt a prompt scaffold from the local Turing Experiments assets Aher et al. [2023]. For each condition, we simulate 15 synthetic participants. Each participant receives offers from 1 to 9 dollars out of a 10-dollar split and decides whether to accept or reject. From these decisions we infer the first accepted offer as the participant’s acceptance threshold, compute a participant-level policy string such as RRAAAAAA, and measure monotonicity violations, policy entropy, and the number of distinct policy profiles.

3.3 Metrics and Statistical Reporting

For SOCIAL IQA, we report accuracy, bootstrap 95% confidence intervals, macro-F1, answer entropy, and pairwise condition comparisons using an exact McNemar-style proxy on discordant items. We also compute Jensen-Shannon divergence between aggregate answer distributions to test whether conditioning changes the overall support.

For the ultimatum task, we report offer-wise acceptance curves, mean acceptance threshold with bootstrap 95% confidence intervals, threshold standard deviation, mean policy entropy, mean mono-

Table 1: SOCIAL IQA local-fidelity results. Higher is better for accuracy and macro-F1. Best values are in **bold**.

Condition	N	Accuracy	95% CI	Macro-F1	Entropy (bits)
Demographic	90	0.700	[0.600, 0.778]	0.701	1.583
Persona	90	0.789	[0.700, 0.867]	0.789	1.583
Unconditioned	90	0.744	[0.644, 0.822]	0.744	1.581

Table 2: Ultimatum-game simulation results. Higher threshold standard deviation and more distinct profiles indicate broader behavioral support, while monotonicity violations should remain low. Best values are in **bold** within each column.

Condition	Participants	Mean threshold	95% CI	Threshold SD	Policy entropy	Distinct profiles
Demographic	15	2.40	[2.13, 2.67]	0.51	0.608	2
Persona	15	2.53	[1.93, 3.00]	1.13	0.590	5
Unconditioned	15	2.67	[2.40, 2.87]	0.49	0.677	2

tonicity violations, and the number of distinct policy profiles. We also report pairwise threshold differences with bootstrap confidence intervals.

3.4 Implementation and Cost

All experiments ran in ‘/workspaces/theory-of-simulation-6163-codex’ with Python 3.12.8 and standard data-analysis libraries, including ‘openai’, ‘datasets’, ‘numpy’, ‘pandas’, ‘scipy’, ‘scikit-learn’, ‘matplotlib’, and ‘seaborn’. The experiments are API-based, so available GPUs were not used. Raw outputs are cached in ‘results/raw/’, summaries in ‘results/summary/’, and figures in ‘figures/’. Total usage was 125,857 tokens: 106,737 input tokens and 19,120 output tokens. Using the GPT-4.1 pricing listed on May 5, 2026, the full study cost approximately \$0.366 OpenAI [2026b].

4 Results

4.1 Social IQA Measures Local Fidelity

Figure 1 reports the aggregate answer counts, and table 1 reports the summary metrics. Persona conditioning achieves the best local fidelity, reaching 0.789 accuracy and 0.789 macro-F1. This is a 4.4-point improvement over the unconditioned baseline and an 8.9-point improvement over demographic conditioning. The paired comparison between persona and demographic prompting yields an exact McNemar-style proxy of $p = 0.0215$, while persona versus unconditioned is smaller and not significant at the 0.05 level ($p = 0.2891$).

The main surprise is that conditioning changes accuracy more than it changes the aggregate answer distribution. Answer entropy is nearly constant across all three conditions at about 1.58 bits, and pairwise Jensen-Shannon divergences are tiny, ranging from 0.0004 to 0.0022. In other words, persona prompts improve which items the model gets right without substantially changing the overall mix of A/B/C selections. This supports local fidelity and some latent-factor responsiveness, but it does not yet show broad distributional change.

4.2 Ultimatum Policies Broaden Slightly but Remain Compressed

Figure 2 shows the acceptance curves, and table 2 reports participant-level summaries. All three conditions produce monotone or nearly monotone policies, which means the simulator behaves consistently as offers improve. However, the thresholds are unrealistically low in every condition. Demographic prompting yields a mean threshold of 2.40, persona prompting 2.53, and unconditioned prompting 2.67. Every condition accepts offers of 3 or more almost universally.

Persona conditioning does widen behavioral support. It increases the threshold standard deviation from about 0.5 to 1.13 and raises the number of distinct policy profiles from 2 to 5. The persona con-

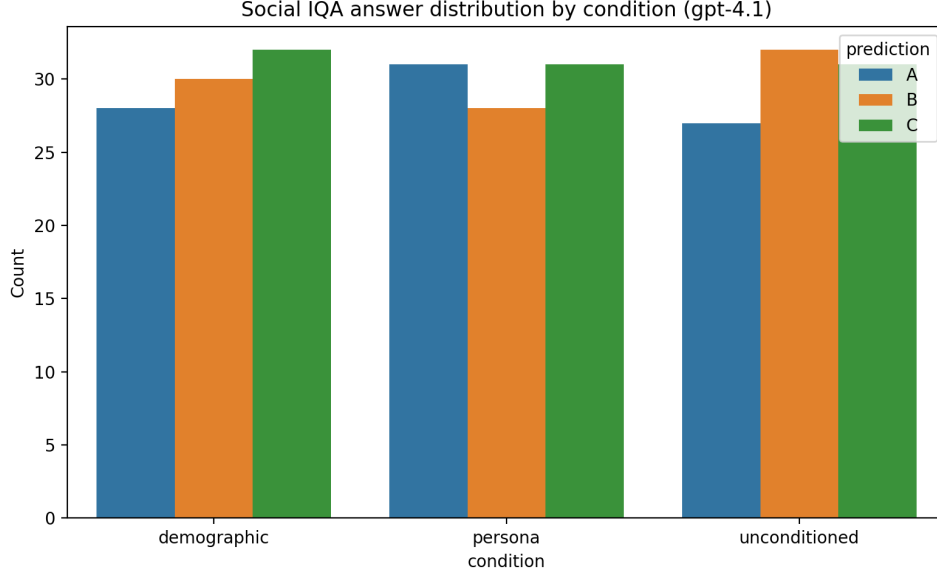


Figure 1: Aggregate SOCIAL IQA answer counts by prompt condition. Persona conditioning improves item-level accuracy, but the aggregate choice distribution remains almost unchanged, consistent with the very small pairwise Jensen-Shannon divergences.

dition is also the only one that produces an always-accept policy and a stricter RRRAAAAA profile. Yet this additional variety still sits within a narrow family of cooperative policies. Even the strictest observed profile begins accepting at offer 3, and one persona accepts every offer from 1 through 9. The model therefore broadens the manifold without recovering human-like tails.

The pairwise threshold intervals reinforce this conclusion. Unconditioned prompting produces a modestly higher mean threshold than demographic prompting, with a difference of 0.267 and a 95% interval of [0.178, 0.351]. By contrast, persona versus demographic and persona versus unconditioned both include zero. Conditioning changes the support more than it changes the central tendency.

4.3 Token Usage

The study is lightweight in monetary terms. SOCIAL IQA used 52,524 tokens, and the ultimatum benchmark used 73,333 tokens, for a combined total of 125,857 tokens. Using the official May 5, 2026 pricing for GPT-4.1, this corresponds to about \$0.151 for SOCIAL IQA, \$0.215 for ultimatum, and \$0.366 overall OpenAI [2026a,b]. The low cost makes this style of evaluation easy to repeat across models, which is useful because simulation validity is likely to change over time.

5 Discussion

What do the results imply for a theory of simulation? The evidence supports a middle position. GPT-4.1 is not just emitting one default answer. Persona conditioning improves SOCIAL IQA accuracy and expands the set of ultimatum policies, which shows some responsiveness to latent factors. At the same time, the added diversity is narrow, structured, and biased toward compliance. This means the model behaves more like a controllable generator of behavioral manifolds than a simulator with full human-outcome coverage.

Surface plausibility, fidelity, and coverage should be evaluated separately. The experiments suggest a practical three-level view. Surface plausibility is easy: the model produces coherent human-like justifications in every condition. Local fidelity is harder but still attainable: on SOCIAL IQA, persona prompting improves performance rather than harming it. Distributional coverage is the bottleneck. The model broadens support from 2 to 5 ultimatum profiles under persona prompting, but

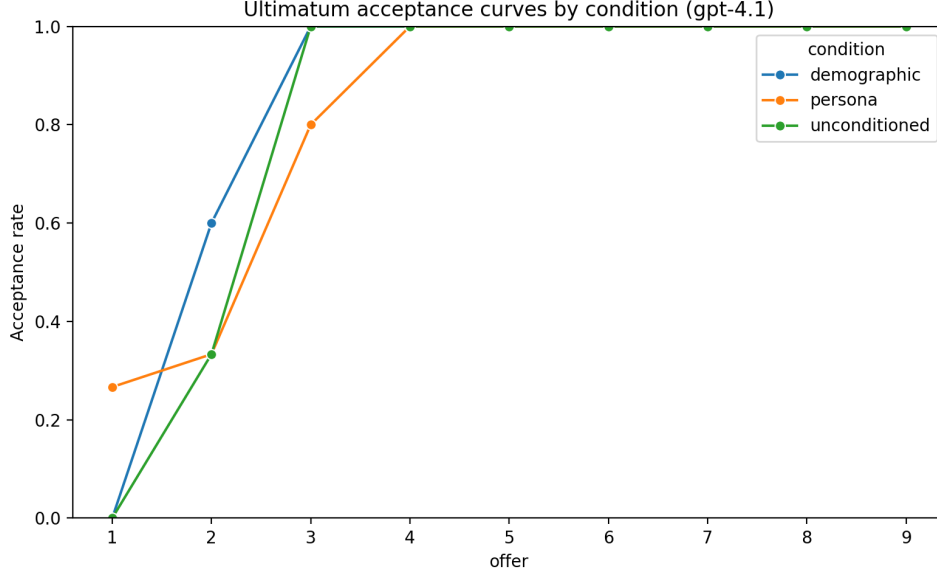


Figure 2: Acceptance curves in the ultimatum game. Persona conditioning produces the widest range of participant policies, but all three conditions accept unfair offers far earlier than expected from human populations, with near-universal acceptance by offer 3 or 4.

those policies remain clustered around very low acceptance thresholds. A theory of simulation that conflates these three levels will overstate current capabilities.

Error analysis clarifies the failure modes. In SOCIAL IQA, the main errors are semantically narrow. The model often chooses a plausible affective state when the benchmark expects a more specific trait label, or it picks a broadly reasonable answer instead of the intended social inference. In the ultimatum task, the failures are structural. Demographic conditioning barely changes the policy family, and persona conditioning mainly introduces profiles that are even more accepting or only slightly stricter. This pattern matches prior concerns about average-person collapse and rationality bias Liu et al. [2025], Chen et al. [2026].

Limitations. Our study tests only one model family and two short-horizon tasks. SOCIAL IQA is likely affected by pretraining exposure, so it is not a pure simulation benchmark. The local OpinionQA mirror lacked the full human-response tables, which prevented a stronger subgroup-alignment analysis. The ultimatum design is lightweight and does not use richer memory or belief-desire-intention scaffolds of the kind explored in prior work Xie et al. [2024], Park et al. [2023]. Finally, the API behavior reflects GPT-4.1 as accessed on May 5, 2026 and may drift over time.

Broader implications. The main operational lesson is that LLMs should not be treated as drop-in substitutes for human populations in settings where subgroup structure or tail risk matters. They are more defensibly used as controllable generators that can help probe hypotheses, prototype interaction policies, or stress-test a bounded region of a behavior space. Stronger claims require separate validation for fidelity, subgroup structure, and tail coverage on the target domain.

6 Conclusion

We studied a simple but consequential question: do modern LLMs simulate a superset of human outcomes once we condition them on latent human factors? Our answer is no, at least not in the strong sense. Using real API calls to GPT-4.1 on a labeled social-judgment benchmark and a Turing-Experiment-style ultimatum game, we find that persona conditioning improves local fidelity and widens the observed policy support, but the resulting behaviors remain compressed within a cooperative and rationalized family.

The main contribution of this paper is therefore conceptual as much as empirical. A useful theory of LLM simulation should separate local fidelity, latent-factor responsiveness, and coverage. Current frontier models can satisfy the first two requirements in some settings, but they still struggle with the third. The result is not failure in an absolute sense; it is a clearer statement of what these systems are good for.

Future work should restore full subgroup-distribution benchmarks such as OpinionQA, compare multiple current model families, and test richer agent architectures on longer-horizon tasks. Those extensions are necessary if the field wants to move from plausible behavioral text generation to defensible human simulation.

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